

Solution Requirements

Date	06 July 2026
Team ID	TMID-SL-2026-01
Project Name	SmartLender - Applicant Credibility Prediction for Loan Approval
Maximum Marks	4 Marks

Functional Requirements (FR) define specific behaviors/features. Non-Functional Requirements (NFR) define how well the system performs.

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority
1	Applicant data capture	System shall collect gender, marital status, dependents, education, employment, home, loan amount/term, o	High
2	Preprocessing	System shall encode categorical fields and scale numeric fields identically to the training pipeline before inference.	High
3	Prediction	System shall load the trained model (rdf.pkl) and return an Approved/Rejected outcome with a confidence score.	High
4	Result presentation	System shall render the prediction and key applicant details on a result page.	Medium
5	Model training pipeline	System shall support retraining on updated data and re-serializing the model/scaler.	Medium
6	Data balancing	System shall balance the training classes using SMOTE before model fitting.	Medium
7	Model evaluation	System shall report train/test accuracy and cross-validation score per model to justify the deployed choice.	Medium
8	Other	System shall be runnable locally via a single command (python app.py).	Low

Step 2: Non-Functional Requirements (NFR)

S.No	NFR Category	Requirement Description	Target Metric
1	Performance & Speed	A single prediction should return quickly enough for interactive use.	≤300 ms per request (local)
2	Scalability	The Flask app should handle a credit officer's typical daily volume of requests without degradation.	Handle 50 concurrent requests without
3	Security & Data Privacy	Applicant data submitted via the form should not be persisted beyond a request.	Do not persist data written to disk by the app
4	Reliability & Availability	The app should start reliably and serve predictions without crashing on valid form submissions.	0 unhandled exceptions on valid form sub
5	Usability & Accessibility	The form should be usable on both desktop and mobile browsers.	Responsive layout down to 375px width
6	Maintainability	Model and scaler should be swappable without changing app.py.	Re-training only touches Training/train_mo